

CELLGUARD · REQUIREMENTS

CellGuard requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-02	Amish Chadha	Status from CGD-CAL-001 v0.6, re-run for the decided responses: R6 and R8 now met on paper; R16 over the value-engineering target by USD 19.15
0.7	2026-10-02	Amish Chadha	R8 restated and R14 relaxed to 0.7 kg; R6, R8 and R14 status after the 2026-10-02 decisions (CGD-DEC-001)
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target; cost wording only, no number changed
0.5	2026-09-30	Amish Chadha	Design for construction (CGD-DDR-003); figures follow CGD-CAL-001 v0.3; no requirement changed status
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3. R1 adds the NMC profile and R11 the SwapCell v0.3 message set (CGD-DDR-001 items 7 and 8,

adopted for TRL 3 pending Amish's review); status from CGD-CAL-001

0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-25	Amish Chadha	Initial scaffold

CellGuard requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs, and will be revised after the lab projects that depend on CellGuard confirm their needs. Status in Table 2 is judged against the TRL 3 calculations in CGD-CAL-001. On paper, 12 of 16 are met; R2, R4 and R7 are at risk, and the estimated cost is over the R16 value-engineering target. On 2026-10-02 Amish decided the responses to R6, R8 and R14 (CGD-DEC-001): R8 and R14 are restated below, and CGD-CAL-001 v0.6 is re-run for them, so R6 and R8 are now met on paper. Changes in v0.3 follow CGD-DDR-001. In v0.4, R16 follows the \$140 budget that Amish set on 2026-09-25, when he accepted the recommendations (CGD-DDR-002); R3 cites the decided protection settings and R9 the decided one-point calibration.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Cover the lab's LFP packs with one board	4 to 16 LFP cells in series (12.8 to 51.2 V nominal, 58.4 V maximum), cell count set in a configuration file. The hardware also supports an NMC profile up to 14 cells (under 60 V), covering SwapCell's 13S pack; LFP firmware is released first	Design review against the front-end data sheet
R2	Measure every cell voltage accurately	± 10 mV at 25 °C and ± 15 mV from –20 to 60 °C, each cell at least every 100 ms	Data sheet review; later comparison with a calibrated meter
R3	Protect the pack from abuse	Cell overvoltage (default 3.65 V), cell undervoltage (default 2.50 V), charge overcurrent, discharge overcurrent, short-circuit trip within 500 μ s, over-temperature, charge lockout below 0 °C; all thresholds in the configuration file; default settings as decided in CGD-CAL-001 Table 2 (SCD 100 mV at 15 μ s, OCD2 40 mV at 20 ms, OCD1 24 mV at 320 ms, OCC 12 mV)	Protection table review; later fault injection on a cell simulator
R4	Stay safe after a single fault	No single failure (microcontroller crash, stuck-on MOSFET, open sense wire, failed sensor) leaves the pack able to be overcharged or over-discharged	Failure mode and effects analysis
R5	Carry the lab's load currents	40 A continuous and 80 A for 10 s at 40 °C ambient, MOSFET junction at 110 °C or less	Thermal calculation; later bench test
R6	Waste little power in the power path	4 W or less on the board at 40 A (MOSFETs, shunt and copper), excluding the external fuse	Loss calculation from data sheets
R7	Balance cells	Passive balancing at 100 mA or more per cell; corrects a 1 % capacity imbalance in 24 h of balancing time on packs up to 100 Ah; board stays at 70 °C or less while balancing	Balance power and time calculation
R8	Give a state of charge the user can trust	Documented method; error within ± 5 percentage points after a full charge, and within ± 10 points during partial cycling between prompted full charges: the firmware prompts for a full charge when about five days pass without one (restated 2026-10-02, CGD-DEC-001)	Algorithm note; later cycling test against a lab reference
R9	Measure current	± 1 % of reading ± 50 mA from –80 to +80 A, with a one-point gain calibration at build	Shunt and front-end error budget
R10	Draw little from an idle pack	300 μ A or less in sleep with outputs on and CAN able to wake; 10 μ A or less in ship mode	Quiescent current budget from data sheets
R11	Talk to host devices openly	CAN 2.0B at 250 kbit/s carrying the SwapCell interface v0.3 message set as a CellGuard profile (subject to the SwapCell project); 3.3 V UART at 115,200 baud for logs and configuration; an enable input that keeps the output off until the host is connected	Interface specification review
R12	Keep a state-of-health log	At least 2,000 charge and discharge event records in non-volatile memory, exportable as CSV; fields sufficient for EU Batteries Regulation state-of-health data	Log format review
R13	Precharge the load	Charge up to 2 mF of load capacitance to 90 % of pack voltage within 1 s before closing the discharge path	Precharge calculation

ID	Requirement	Target	Verification (TRL 3 or later)
R14	Be small and light	230 x 120 x 40 mm or smaller, 0.7 kg or less without power cables (relaxed from 0.6 kg on 2026-10-02, CGD-DEC-001)	Massing model; later weighing
R15	Be buildable and open	No BGA or leadless parts finer than 0.5 mm pitch; assembly with a hot plate or hot-air station; hardware CERN-OHL-S-2.0, firmware MIT (reused Apache-2.0 files keep their own license in <code>firmware/third_party/</code>)	Design review of the parts list
R16	Stay within the value-engineering target	\$140 or less in parts at quantity one, excluding cells (a hypothetical control target, not a limit; raised from \$120, CGD-DDR-002)	Priced BOM

Table 2. Status at TRL 3 (CGD-CAL-001; calculations, nothing measured).

ID	Status	Basis
R16	Over the value-engineering target by \$19.15	Value-engineering target: USD 140.00. Estimated cost of the constructable design: USD 159.15 (USD 19.15 over the target), mainly the 1.5 mΩ-class switches decided on 2026-10-02 (\$20.00 more than the 2.5 mΩ parts)
R2	At risk	Total error ± 15 mV from -40 to 85 °C meets the range target; at 25 °C the data sheet gives only a typical figure under 10 mV
R4	At risk	With the BQ77216-class secondary protector and SCP fuse every single fault in the failure analysis ends safe; an SCP fuse rated for 40 A at 60 V DC is not yet confirmed
R7	At risk	103 mA; 9.7 h for 1 % on 100 Ah; PCB 56.4 °C but a balance resistor hotspot about 73.9 °C at 40 °C ambient
R1	Met	12.8 to 58.4 V; 100 V MOSFETs with 1.71 times margin; front end rated 3 to 16 cells
R3	Met on paper	Thresholds inside the front-end ranges; short-circuit trip within 25 μ s
R5	Met on paper	Junction 51.0 °C at 40 A (top-cooled package), 77.6 °C bound at 80 A for 10 s (TOLL)
R6	Met on paper (was not met at v0.7)	3.47 W on the board at 40 A with the 1.5 mΩ-class switches decided on 2026-10-02, of which 0.64 W is the secondary protector's SCP fuse; about 4.4 W with the former 2.5 mΩ parts
R8	Met on paper with calibration (was not met at v0.7)	5.0 points after a full charge; 8.7 points on 20 Ah after five days between prompted full charges with a calibrated coulomb counter offset (24.5 uncalibrated), 6.7 on 100 Ah. The calibrated offset is TI's typical figure; the 4S 20 Ah check decided on 2026-10-02 tests it
R9	Met on paper with calibration	0.404 A error at 80 A after a one-point gain calibration, against 0.850 A
R10	Met on estimate	54.5 μ A in sleep, 6.0 μ A in ship mode
R11	Met by design	SwapCell v0.3 message set; 1.84 % bus load at 250 kbit/s
R12	Met	65,536 records of 32 bytes in 2 MiB
R13	Met	90 % in 0.46 s; 3.38 J
R14	Met on paper	220 x 110 x 33 mm meets the size; 0.664 kg against the 0.7 kg target relaxed on 2026-10-02 (CGD-DEC-001)
R15	Met by design review	TQFP-48 at 0.5 mm, TSSOP-24 at 0.65 mm, no BGA

■ Assumptions

- LFP cell limits are typical defaults (3.65 V maximum and 2.50 V minimum per cell; charge 0 to 45 °C; discharge -20 to 60 °C). Every build must take its limits from the actual cell data sheet.
- Reference packs: 4S 280 Ah (12.8 V, about 3.6 kWh) for storage and 16S 20 Ah (51.2 V, about 1 kWh) for vehicles.
- The 40 A rating covers a 400 W inverter on a 12.8 V pack (about 31 A) and a 30 A vehicle controller. Larger loads need a larger board or a contactor, which is out of scope.
- Prices are indicative single-unit prices from distributors in September 2026 and will change.

- `budget_usd` in `project.yaml` is the \$140 value-engineering target, raised from \$120 by Amish on 2026-09-25 (CGD-DDR-002); R16 is reported against the \$140 value-engineering target.
- Responses to R6, R8 and R14, decided by Amish on 2026-10-02 (CGD-DEC-001): 1.5 mΩ-class switches for R6; a full-charge prompt after about five days without one, with R8's partial-cycling clause restated to apply between prompted full charges; R14 relaxed to 0.7 kg with the 4 mm plate kept.